Ruşen Birben

AI Researcher & Engineer

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AI and Data Engineering graduate from Istanbul Technical University and independent researcher with METU's Applied NLP Group. Research interests center on LLM reliability and interpretability, with experience in Turkish NLP, machine unlearning, and LLM applications for finance.

EDUCATION

B.Sc. AI and Data Engineering
2022 - 2025

Istanbul Technical University
GPA: 3.61 / 4.00

Graduated at 09.07.2025

Electronics and Communication Engineering
2020 - 2022

Istanbul Technical University

Transferred to AI & Data Engineering

SCHOLARSHIPS & AWARDS

1416 YLSY Scholarship · M.Sc. & Ph.D. 2026

Awarded government funding for overseas master's study followed by doctoral study, covering tuition fees, a monthly living stipend, health insurance, travel, language-exam fees, and study supplies. Following completion of the funded studies, will return to Türkiye as an assistant professor.

1st Place · Agenthon 2026, Track 3 (Simulation) 2026

Ranked 1st of 55 teams in the simulation track, building as a team an ABIDES-compatible market simulator that maximizes event throughput while preserving matching-engine semantics. Selected to present a team poster at the Agenthon workshop at NeurIPS 2026.

Kaggle Bronze Medal · The Pokémon Company PTCG AI Battle Challenge 2026

Ranked 677th of 6,807 teams (top 10%) in Kaggle's simulation competition, developing and evaluating partially observable game-playing agents with replay-based behavioral cloning, belief-aware Monte Carlo planning, and self-play.

RESEARCH EXPERIENCE

Independent Researcher
03/2026 - Present

METU Applied NLP Group
Ankara, TR

- Co-organize the SIGTURK 2027 Shared Task at the SIGTURK 2027 Workshop (ACL 2027) with Çağrı Toraman, Evren Ayberk Munis, and İlker Kesen. Contribute to data creation and evaluation methodology; built the shared-task website and evaluation pipeline.
- Support GemmaTurk through training-data collection, synthetic instruction and reasoning data generation for mid-training, and pretraining experiments.
- Lead METU NLP Connect, producing expert interviews.

AIOps Research Intern
08/2024 - 09/2024

Havelsan
Istanbul, TR

- Investigated software-log analysis for predictive maintenance; developed prototypes using HDFS logs.

NLP Research Intern
07/2024 - 08/2024

ITU NLP Lab
Istanbul, TR

- Researched financial LLM applications involving sentiment analysis, tool use, reasoning, forecasting, and retrieval-augmented generation. Contributed experiments to the Kaira undergraduate capstone.

PUBLICATIONS & RESEARCH CONTRIBUTIONS

Exact First, Then Fast: An Event-for-Event C++ Rebuild of the Agenthon ABIDES Market 2026

Agenthon Workshop @ NeurIPS 2026 - Accepted for in-person poster presentation Competition

- Göktürk Batın Dervişoğlu, Muhammed Faruk Aytin, Muhammed Rüşen Birben.
- Rebuilt the Agenthon ABIDES market simulator in C++, preserving event ordering, random-number sequences, and reference outputs. Validated exact agreement across 6,900 randomized comparisons and achieved a 1,299× simulation-loop speedup on the largest public benchmark, measured on the same laptop.
- Atlanta, USA, December 12, 2026.

Turkish Cultural Knowledge Dataset 2026

MRL Workshop @ EMNLP 2026 - Accepted dataset contribution; joint benchmark paper in preparation

Shared Task

- Led the source-grounded agent pipeline and data preparation for a Turkish cultural knowledge dataset spanning 15 domains.
- Screened 1,200 candidate question-answer pairs; selected 500 for sociological review by Büşra Tufan and prepared the final 300-pair submission.
- Contributed to English translation review under the research guidance of Çağrı Toraman.

SELECTED PROJECTS

Kaira 2025

Undergraduate Capstone: Financial Reasoning LLM

- Fine-tuned Qwen-based LLMs with GRPO, Unsloth, and LoRA; ran FinQA and GSM8K experiments on reward design and financial reasoning. Developed an Ollama-based chat application.

Tobor 2024 - 2025

Autonomous Warehouse Robot Code Video

- Implemented the box pickup/drop-off mechanism in a team-built ROS2 warehouse robot with navigation and YOLO perception; shared data annotation and supported other algorithms.

AIZheimer 2024

Machine Unlearning in Stable Diffusion Code

- Adapted Forget-Me-Not-based unlearning for Stable Diffusion 2.1. Implemented attention controllers and explored loss objectives; qualitatively assessed concept removal and retention of unrelated generation capabilities.

Biting The Bytes 2024

Turkish Diacritic Restoration Code Demo

- Led ByT5-small training and hyperparameter tuning with PEFT/LoRA for byte-level Turkish diacritic restoration. Shared debugging and Hugging Face Spaces deployment with a teammate.

To-AI-or-Not-to-AI 2023

GPT Detector Code Demo

- Built an ensemble of three fine-tuned RoBERTa detectors combined by majority vote, reaching 94.21% accuracy on the held-out test split of a 26,270-example dataset prepared with teammates.

PROFESSIONAL EXPERIENCE

Co-Founder & Technical Lead
08/2025 - 04/2026

Fiction Studios Ankara, TR

- Co-founded Fiction Studios at Bilkent Cyberpark; led technical development and coordinated three additional contributors. Built a multilingual LLM-powered museum and tourism guide, alongside customer and investor discussions.

Junior AI Engineer
03/2025 - 07/2025

CyberQuote Istanbul, TR

- Built RAG pipelines that retrieve up-to-date context from fragmented internal sources and PDF financial reports, and agent pipelines for internal financial LLM tools. Also developed news-sentiment pipelines and web integrations at a Phillip Capital subsidiary.

TECHNICAL SKILLS

Languages Python / JavaScript/TypeScript / SQL / C/C++

Frameworks & Tools PyTorch / TensorFlow / Transformers / PEFT / LoRA / Unsloth / Scikit-learn / Pandas / NumPy / ROS2 / YOLO

Specialties NLP / LLMs / RAG / Machine Unlearning / Model Evaluation

COURSES / LANGUAGES

Deep Learning Specialization DeepLearning.AI · Coursera

AWS Cloud Technical Essentials AWS · Coursera

Turkish Native **English** TOEFL iBT 109/120 (November 8, 2025)

Interests: Coffee / Books / Photography / Anime & Manga